

liebreadth

Breadth in finite dimensional algebras

1.0

24 September 2026

Óscar Fernández Ayala

Óscar Fernández Ayala Email: oscar00ayala@gmail.com

Homepage: <https://Osferay.github.io>

Contents

1	Preface	3
1.1	Introduction	3
2	Lie algebras and breadth	4
2.1	Attributes	4
2.2	Breadth	5
2.3	Inflation of Lie algebras	6
	References	8
	Index	9

Chapter 1

Preface

In this package we compute the breadth of Lie algebras. The details of the algorithms are available in [EF26]. This package also includes functions to compute covered Lie algebras of maximal class using a process called inflation, which is described by Caranti, Mattarei, and Newman [CMN97].

1.1 Introduction

Given a finite-dimensional algebra L , we define its lower central series as $L = L^1 > L^2 > \dots$, where $L^{i+1} = L^i L$. The algebra L is nilpotent if there exists $c \in \mathbb{N}$ such that $L^{c+1} = \{0\}$, and the minimal such c is called the class $\text{cl}(L)$ of L . For a nilpotent algebra L , the type of L is the vector $(d_1, \dots, d_c)$, where $d_i = \dim L^i$. A nilpotent algebra L is said to be of maximal class if the type of L is $(2, 1, \dots, 1)$. The centralizer of $x \in L$ is the subspace of L defined by $C_L(x) = \{a \in L \mid ax = 0\}$. For an algebra L , we define

$$\text{br}(L) = \max\{\text{br}(x) \mid x \in L\}, \quad \text{where } \text{br}(x) = \dim(L) - \dim(C_L(x)).$$

The class-breadth conjecture asserts that $\text{cl}(L) \leq \text{br}(L) + 1$ for an algebra L . The conjecture holds for nilpotent Lie algebras over infinite fields and for nilpotent associative algebras over arbitrary fields.

Let L be a Lie algebra over a field K , and let $B = \{b_1, \dots, b_n\}$ be a basis of L . The multiplication in L is described by structure constants defined by

$$b_i \cdot b_j = \sum_{k=1}^n c_{ijk} b_k \quad \text{for } 1 \leq i, j \leq n.$$

For each k , write $C_k = (c_{ijk})_{1 \leq i, j \leq n}$ for the $n \times n$ matrix over K with entries c_{ijk} , and these matrices are the structure matrices of L . For $x = x_1 b_1 + \dots + x_n b_n \in L$, let $\bar{x} = (x_1, \dots, x_n) \in K^n$ denote the coefficient vector of x . Then $C_k \bar{x}^{tr}$ is a column vector over K . We write $M_B(x)$ for the $n \times n$ matrix over K whose k -th column is $C_k \bar{x}^{tr}$, which is the adjoint matrix of L with respect to the basis B .

Chapter 2

Lie algebras and breadth

2.1 Attributes

2.1.1 LieClass (for IsLieNilpotent)

▷ `LieClass( $L$ )` (attribute)

Computes the class of the nilpotent Lie algebra L .

2.1.2 LieType (for IsLieNilpotent)

▷ `LieType( $L$ )` (attribute)

Computes the type of the nilpotent Lie algebra L .

2.1.3 IsOfMaximalClass (for IsLieNilpotent)

▷ `IsOfMaximalClass( $L$ )` (property)

Returns: true or false

Returns whether the nilpotent Lie algebra L is of maximal class or not.

2.1.4 StructureMatrices (for IsLieAlgebra)

▷ `StructureMatrices( $L$ )` (attribute)

Computes the structure matrices of the nilpotent Lie algebra L .

2.1.5 BasisLieCenter (for IsLieAlgebra)

▷ `BasisLieCenter( $L$ )` (attribute)

Returns the basis elements of the Lie algebra L that are contained in the center of L .

2.1.6 BasisLieDerived (for IsLieAlgebra)

▷ BasisLieDerived(L) (attribute)

Returns the basis elements of the Lie algebra L that are contained in the derived subalgebra of L .

2.1.7 LieAdjointMatrix (for IsLieAlgebra)

▷ LieAdjointMatrix(L) (attribute)

Computes the adjoint matrix of the Lie algebra L .

2.1.8 PrintLiePresentation (for IsLieAlgebra)

▷ PrintLiePresentation(L) (attribute)

Prints the Lie presentation of the Lie algebra L .

2.2 Breadth

2.2.1 InfoLieBreadth

▷ InfoLieBreadth (info class)

Info class for the functions of the breadth of Lie algebras.

2.2.2 LieBreadth (for IsLieNilpotent)

▷ LieBreadth(L) (attribute)

Computes the breadth of the Lie algebra L .

2.2.3 IsTrueClassBreadth (for IsLieNilpotent)

▷ IsTrueClassBreadth(L) (property)

Returns: true or false

Returns whether the class-breadth conjecture holds for the Lie algebra L .

2.2.4 TGroupBreadth

▷ TGroupBreadth(G) (function)

Computes the breadth of the $\mathcal{T}$ -group G .

2.3 Inflation of Lie algebras

A grading for a Lie algebra L is a decomposition $L = \bigoplus_{i=1}^n L_i$ that respects the Lie bracket, *i.e.* $[L_i, L_j] \subseteq L_{i+j}$. All nilpotent Lie algebras can be graded by taking $L_i = \gamma_i(L)/\gamma_{i+1}(L)$. Let L be a nilpotent Lie algebra of maximal class. The two-step centralizers are the sets $C_i = C_{L_1}(L_i) = \{x \in L_1 \mid [x, L_i] = 0\}$ for all $2 \leq i \leq c$. Let $\mathcal{C} = \{C_i\} \setminus L_1$. We say that a Lie algebra of maximal class is covered if the set $\mathcal{C}$ consist of all one-dimensional subspaces of L_1 .

Let $L = \bigoplus_{i=1}^n L_i$ be a graded Lie algebra over $K = \mathbb{F}_q$ for some prime power $q = p^n$. Consider the field extension $A = K[\varepsilon]/\langle \varepsilon^p \rangle$ which is a vector space over K of dimension p . The algebra A is an associative, commutative algebra with a unit. The algebra $L \otimes A$ over K , defined by $[x \otimes a, y \otimes b] = [x, y] \otimes ab$, is a graded Lie algebra. Let M be a maximal ideal of L and consider the Lie subalgebra $M^\uparrow = M \otimes A$. Let D be a derivation of L of degree 1. Define $D^\uparrow$ by: $D^\uparrow(x \otimes \varepsilon^i) = D(x) \otimes \varepsilon^i$, which is a derivation of $M^\uparrow$ of degree 1. Define the derivation $E \in \text{Der} M^\uparrow$ by $E(x \otimes \varepsilon^i) = D^\uparrow(x \otimes \varepsilon^i \cdot \varepsilon^{p-1}) + 1 \otimes \partial_\varepsilon(\varepsilon^i)$. For $s \in L_1 \setminus M$, take $D = \text{ad}_s$, and extend it naturally to $M^\uparrow$. Denote $E_{s'}$ the derivation defined above. The *inflation* $^M L$ of L at M by $s \in L_1 \setminus M$ is the graded Lie algebra obtained as an extension of $M^\uparrow$ by an element s' , which is the extension of s that induces the derivation $E_{s'}$, that is,

$$[x \otimes a, s'] = E_{s'}(x \otimes a) = \text{ad}_{s'}(x \otimes \varepsilon^i) + x \otimes \partial_\varepsilon(\varepsilon^i).$$

2.3.1 InfoInflation

▷ InfoInflation

(info class)

Info class for the functions of the inflation of Lie algebras.

2.3.2 LieNilpotentGrading (for IsLieNilpotent)

▷ LieNilpotentGrading(L)

(attribute)

Computes a grading of the nilpotent Lie algebra L using the lower central series.

2.3.3 LieTwoStepCentralizers (for IsLieNilpotent)

▷ LieTwoStepCentralizers(L)

(attribute)

Computes the two-step centralizers of the Lie algebra L .

2.3.4 IsLieCovered (for IsLieNilpotent)

▷ IsLieCovered(L)

(property)

Returns: true or false

Returns whether the Lie algebra L is covered or not.

2.3.5 PolynomialAlgebra (for IsField and IsFinite)

▷ PolynomialAlgebra(F)

(attribute)

For a finite field F , computes the polynomial algebra $F[x]$.

2.3.6 LieCoveredInflated (for IsInt)

▷ LieCoveredInflated(n) (attribute)

For $n = 2, 3, 4$, computes the covered Lie algebras of maximal class using the polynomial algebra of dimension n . For $n = 3$, one can directly load the Lie algebra by reading the file `\textrm{inflation\_3.g}`.

2.3.7 LieMinimalQuotientClassBreadth (for IsLieAlgebra)

▷ LieMinimalQuotientClassBreadth(L) (attribute)

Given a covered Lie algebras of maximal class L , computes the minimal quotient for which the class-breadth conjecture does not hold.

References

- [CMN97] A. Caranti, S. Mattarei, and M. F. Newman. Graded lie algebras of maximal class. *Transactions of the American Mathematical Society*, 349:4021--4051, 1997. [3](#)
- [EF26] B. Eick and Ó. Fernández Ayala. On the class-breadth conjecture for algebras and t -groups. *Journal Algebra*, 703:65--76, 2026. [3](#)

Index

BasisLieCenter
 for IsLieAlgebra, [4](#)
BasisLieDerived
 for IsLieAlgebra, [5](#)

InfoInflation, [6](#)
InfoLieBreadth, [5](#)
IsLieCovered
 for IsLieNilpotent, [6](#)
IsOfMaximalClass
 for IsLieNilpotent, [4](#)
IsTrueClassBreadth
 for IsLieNilpotent, [5](#)

LieAdjointMatrix
 for IsLieAlgebra, [5](#)
LieBreadth
 for IsLieNilpotent, [5](#)
LieClass
 for IsLieNilpotent, [4](#)
LieCoveredInflated
 for IsInt, [7](#)
LieMinimalQuotientClassBreadth
 for IsLieAlgebra, [7](#)
LieNilpotentGrading
 for IsLieNilpotent, [6](#)
LieTwoStepCentralizers
 for IsLieNilpotent, [6](#)
LieType
 for IsLieNilpotent, [4](#)

PolynomialAlgebra
 for IsField and IsFinite, [6](#)
PrintLiePresentation
 for IsLieAlgebra, [5](#)

StructureMatrices
 for IsLieAlgebra, [4](#)

TGroupBreadth, [5](#)